Engineering Mathematics - I · Unit_05 · MATLAB

DRAFT — UE26MA141A · Unit_05 · MATLAB · Plotting of Curves (2D and 3D), Plot Labeling, Curve Labeling

Standalone video lesson · authored study notes with source citations

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· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

Plotting of Curves (2D and 3D), Plot Labeling, Curve Labeling

1 Plotting of Curves (2D and 3D), Plot Labeling, Curve Labeling

Study focus: Plotting of Curves (2D and 3D), Plot Labeling, Curve Labeling. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment.

EV0000062 · EV0000063

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

I am Professor Nagesh Ham, faculty member Department of Science and Humanities.

Key details

- This is the fourth session of MATLAB.
- In this session we learn plotting using MATLAB.
- There are three cases.
- Case number one, Cartesian curves that is y equal to f of x.
- Case number two polar curves that is R equal to F of theta.
- Case number three parametric curves x equal to f of t and y equal to g of t.
- To plot the graph of a function, we need to take the following steps.
- We first define the variable x, then specify the range of values for the variable x.
- We define the function y equal to f of x.
- Finally we use plot command in order to plot the Cartesian curve that is y equal to f of x.
- The plot command is used to create two dimensional plots.
- Before we plot this graph, we have to use the first one.
- Then as you can see, we have to have a 1.
- this is a three crossary matrix.
- And this is a normal or usual matrix multiplication what we know A star b.
- But if you look at A dot star B.
- As you can see the first element of this A is multiplied with this 4.
- Then similarly, multiplication of 2 and 5, so the corresponding number is 10.

- This is how the dot operator works.
- The dot operator is used with multiplication, division and exponentiation to create element wise operation.
- That's what we see here.
- If you look at A star b this is a usual matrix multiplication what we know, right?
- I repeat, the dot operator is used with multiplication, division, exponentiation to create element wise operation.
- And we know this is a procedure.
- We first define the variable x and then we specify the range of values for the variable x.
- Then we define the function y equal to f of x for which we need to plot the graph.
- Then finally we use plot of x comma y command.
- Let x equals two so minus 100.
- This is colon.
- It means x starts with minus 2, it goes up to 4 with a step size 0.01.
- Then define the function y equal to.

Example or application from the lecture

- Let's start with the Cartesian case that is y equal to f of x.
- In order to understand this concept, let us look at the example y equal to 3.5 power 0.5 x into cos of 6x where x lies between minus 2 and plus 4.
- Let's understand the concept of dot operator in Matlab.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000062 · EV0000063 · EV0000064 · +1 source units in pack.json

Core explanation · 05:00–09:59

three point five dot this is a dot operator remember cap minus zero point five x again this has to be star x then dot operator star cos of six times x right then finally use the command plot of x comma y.

Key details

- What happens if you give plot of y comma x.
- This is shape of the graph.
- Way come my X.
- This is the shape of the curve.
- Is it clear?
- This is the error what we get.
- Is it clear?
- y equal to 3.5 powers minus 0.5 x into cos of 6x.
- Square then plot of X comma y.
- Then plot of x comma y.
- When we decrease the increment, the graph becomes smoother.
- When we decrease the increment, the graph becomes smoother.
- That's it.

- Is it clear?
- We know that.
- A function of two independent random variables defines a surface in three dimensional space.
- Step number one.

Example or application from the lecture

- For example, let me cancel this, let me delete this, right?
- So let us take another example y equal to f of x square.
- So in this example we will draw two graphs with the same function, but in the second time we will reduce the value of increment.

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EV0000066 · EV0000067 · EV0000068 · +1 source units in pack.json

Core explanation · 10:00–14:59

We first create a set of points over the domain of the function using mesh grid command.

Key details

- Next we assign the function itself.
- Finally we use a command called surf.
- to plot the surface.
- clear the screen we can write it is x y you can also give capital x y there mesh grid mesh grid of minus two right colon minus point two that is the step size then colon you can give two.
- You start from minus two it goes up to four right then you c call the command z equal to the function is x into x dot product exp for exponential minus x square minus x square means dot cap square minus y dot cap square.
- Finally we use a command surf of x comma y comma z right.
- Is it clear?
- A function of two independent random variables defines a surface in three-dimensional space.
- You can rotate however you want.
- Is it clear?
- See this how you can rotate.
- Is it clear?
- Using MATLAB we can add title labels along the x axis as well as y axis of the graph.
- Commands generate labels along X axis as well as Y axis.
- Finally the title command allows you to put a title on the graph.
- Y label you can say s single code sign of X right then you can give title so title it is.
- Sine of X graph.
- Is it clear?
- And then you can see the corresponding answer is going to be the corresponding graph is this.
- As you can see X axis labeled by X.

Example or application from the lecture

- So in order to understand this concept let us take a simple example let us plot the surface for z equal to x into e power minus of x square plus y square.
- Said let's look at the illustration.
- Okay, let us take a simple example another example you can take sine x cos y.

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EV0000070 · EV0000071 · EV0000072 · +1 source units in pack.json

Core explanation · 15:00–19:59

Y axis is labeled by sine of x and this is the name of the graph.

Key details

- Sine of x graph.
- Is it clear?
- And this is the corresponding answer.
- 0 starting from 0, increment value 0.01 colon 2 pi 2 star pi right.
- Then define the function by rho.
- Rho equal to we are looking for sine 2 theta sine of 2 star theta right multiplication dot operator should be there then cost of two star theta is it clear then use polar plot command polar plot it is simply theta comma row is clear so this theta gamma row.
- Is it clear?
- Then polar plot.
- And this is a graph.
- Cas2 star theta here.
- And then copy paste it here.
- And this is a graph of cos2 theta looks like.
- Is it clear?
- Sims function creates a variable dynamically.
- Is it clear?
- In order to plot parametric curve okay before that what is the equation of a parametric curve?

Example or application from the lecture

- the equation of a polar curve is given by r equal to f of theta clear so in order to plot a polar curve we use this command polar plot of theta comma rho here theta is the angle in radians and rho is a radius value for each point.
- Let us take a simple example r equal to sine 2 theta into cos 2 theta right.
- Let's look at another example, let us take you can take r equal to one minus cos theta, you can take some example.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000074 · EV0000075 · EV0000076 · +1 source units in pack.json

Core explanation · 20:00–24:59

Is it clear?

Key details

- In order to plot this we use this simple command.
- Then we use x t equal to sine t means sin of t.
- colon then y t equal to cos of t then Z t equal to T right so finally we use F plot three X T Y T Z T clear So this is the shape of the graph what we get.
- Is it clear?
- You just ignore these three.
- You get an error.
- Is it clear?
- Therefore, we need to use f plot 3 x t y t and z t.
- parametric curve x equal to e power minus t by 10 sin phi t y equal to e power minus t by 10 cos phi t z equal to t over the interval minus y and plus minus 10 to plus 10.
- Is it clear?

Example or application from the lecture

- So it is other than minus y to plus phi we use another command which we shall see in the next example.
- Take an example let us plot a parametric curve x equal to sin t y equal to cos t z equal to t over the default parameter minus y2 plus y.
- Suppose intentionally we make a mistake.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000078 · EV0000079 · EV0000080 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Plotting of Curves (2D and 3D), Plot Labeling, Curve Labeling transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/AV_Summaries/004_Plotting_of_Curves_2D_and_3D_Plot_Labeling_Curve_Labeling__e96c8141e2597d5f.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Plotting of Curves (2D and 3D), Plot Labeling, Curve Labeling transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/AV_Summaries/004_Plotting_of_Curves_2D_and_3D_Plot_Labeling_Curve_Labeling__e96c8141e2597d5f.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Plotting of Curves (2D and 3D), Plot Labeling, Curve Labeling transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/AV_Summaries/004_Plotting_of_Curves_2D_and_3D_Plot_Labeling_Curve_Labeling__e96c8141e2597d5f.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9982; only the JSON segmentation supplies the displayed timestamps.
Plotting of Curves (2D and 3D), Plot Labeling, Curve Labeling transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/AV_Summaries/004_Plotting_of_Curves_2D_and_3D_Plot_Labeling_Curve_Labeling__e96c8141e2597d5f.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 4 rendered pages in 001_Slides__7ac00f797de85ee4.slides.md have no readable `.slides.md` text; their original page images are included.
2. 2 rendered pages in 002_Slides__ef919841425356a4.slides.md have no readable `.slides.md` text; their original page images are included.
3. 1 rendered pages in 003_Slides__3d7af4a88d5b4490.slides.md have no readable `.slides.md` text; their original page images are included.
4. 1 rendered pages in 004_Slides__4150cd4c33be9a2c.slides.md have no readable `.slides.md` text; their original page images are included.
5. No historical archive question page met the unit-match candidate threshold; all subject-level and ambiguous pages remain in datasets/question-archive.json.
6. 2 video-to-slide associations were not promoted to a match. Candidate rankings are listed in the transcript corpus and video-slide-matches.csv; the slide decks appear separately.
7. 4 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.

8. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
9. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
10. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
11. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00014	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0057. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00015	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0057. Sidecar variants compare...
SRC00016	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0057. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00017	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0057. Sidecar variants compare...

C Revision and retrieval practice